

Epidemiology

Inter- and intra-ethnic differences for female breast carcinoma incidence in the continental United States and in the state of Hawaii

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Summary

Background. Ethnic diversity is well-documented for female breast carcinoma incidence in the continental US but is not so well-established in the state of Hawaii.

Methods. Using the Surveillance, Epidemiology, and End Results (SEER) program, we analyzed $n=323,607$ *in situ* and invasive female breast cancer cases for major ethnic groups in the continental US and in Hawaii, diagnosed during the years 1992–2002.

Results. In the continental US, age-specific incidence rate patterns and prognostic factor profiles were good-risk for Asian or Pacific Islanders (API), intermediate for Whites, and poor-risk for Blacks. For example, early age-at-onset, high nuclear grade, aggressive histopathologic subtypes, and hormone receptor negative expression was associated with Black race in the continental US. In Hawaii, age-specific rate and prognostic profiles were more favorable for API than for White women, albeit not so striking as in the continental US.

Conclusion. We observed *inter-* and *intra-*ethnic differences for female breast carcinoma in the continental US and in the state of Hawaii. While *inter-racial* disparities were expected, *intra-racial* differences were somewhat unexpected and possibly due to variations in racial subgroup mixing and/or cultural assimilation. For example, API women with breast carcinoma in the continental US included 96.03% Asians and 2.4% Pacific Islanders. In contrast, API women with breast carcinoma in Hawaii included 76.52% Asians and 23.46% Pacific Islanders. Moreover, APIs were more likely to be first-generation migrants in the continental US ($\approx 92\%$) than in Hawaii ($\approx 34\%$). Future studies should attempt to disaggregate racial data to separately characterize epidemiological patterns for individual ethnic groups.

Introduction

In the continental US and in the state of Hawaii, breast carcinoma is the most common tumor diagnosed in women and is the second-leading cause of cancer death in females [1]. Previous national studies have shown racial differences in breast carcinoma incidence and prognosis, including the tendency for certain non-White minority women to be diagnosed with more advanced stages of breast carcinoma and to experience lower rates of survival after diagnosis [2,3].

While trends in ethnic breast carcinoma disparity are well-established for women in the continental US, fewer studies have examined population-based data specific to the state of Hawaii. Hawaii also has fewer Hispanic Whites as well as higher concentrations of several Asian and Pacific Islander subgroups; and therefore, it may be

useful to compare populations in Hawaii with those observed in the continental US.

Specifically, this study described ethnic-related patterns in breast carcinoma incidence and survival for the following groups; White, Black, and API women in the continental US as well as White and API women in Hawaii. For each ethnic group, breast carcinoma incidence was described according to standard demographics, tumor characteristics, and age distribution at diagnosis (rates and frequency). Actuarial survival and mortality hazard function also were compared between the groups as a metric of outcome data, providing insight into biologic phenomenon, efficacy of cancer treatment, and post-treatment surveillance. Additionally, we examined racial subgroup composition and place of birth for each major ethnic group.

Materials and Methods

We utilized the National Cancer Institute's (NCI's) SEER Cancer Incidence Public-Use 13 Registry Database (November 2004 submission) to analyze breast carcinoma cases, diagnosed during the years 1992 to 2002 [4]. One record each for 323,607 female *in situ* and invasive breast cases was associated with population data for White, Black, American Indian/Alaska Native (AI/AN), and Asian or Pacific Islanders (API). We stratified the major ethnic groups from the combined registries in the continental US and the one registry in the state of Hawaii. Ethnic groups in the continental US were White ($n=265,733$), Black ($n=27,412$), and API ($n=16,518$). Ethnic groups in Hawaii were White ($n=2995$) and API ($n=7390$). There were not enough Black ($n=56$) or AI/AN ($n=28$) women in Hawaii for analysis, Table 2.

Data on standard demographics and tumor characteristics related to the American Joint Committee for Cancer (AJCC) staging system [5] (i.e., tumor size, axillary lymph node status) as well as characteristics related to biologic features (i.e., grade, histopathology, hormone receptor expression) were dichotomized into low-risk and high-risk groups. For example, tumor size was categorized as ≤ 2.0 cm (low-risk) versus > 2.0 cm (high-risk); lymph node status was negative (low-risk) versus positive (high-risk); tumor grade was low (grades 1–2) versus high (grades 3–4); estrogen receptor (ER) and progesterone receptor (PR) expression were categorized separately as positive (low-risk) or negative (high-risk).

Histopathologic subtypes were identified using codes from the International Classification of Diseases for Oncology-3rd edition (ICD-O-3) [6–8]; i.e., duct carcinoma not otherwise specified (Duct NOS, ICD-O-3 code 8500), tubular carcinoma (ICD-O-3 codes 8211), lobular carcinoma (ICD-O-3 codes 8520), medullary carcinoma (ICD-O-3 codes 8510), inflammatory carcinoma (ICD-O-3 code 8530), papillary carcinoma (ICD-O-3 codes 8050, 8260, 8503), mucinous carcinoma (ICD-O-3 codes 8480), and Paget's disease (ICD-O-3 codes 8540–8543). Other, missing, or unknown data were classified as 'other or unknown'.

Incidence

Age-adjusted incidence rates (2000 US standard population) were calculated using SEER *Stat 6.1.4 and expressed per 100,000 woman-years. Relative risks for tumor characteristics were expressed as incidence rate ratios (RR) with 95% confidence intervals, where the incidence rate for a given racial characteristic was compared to a referent characteristic among White women in the continental US with an assigned RR of 1.0. Using SPLUS 6.2, age-specific incidence rate curves by 5-year age increments were plotted on a log–log scale [9,10]. The density function in SPLUS 6.2, which utilizes a smoothing method of the corresponding age-at-diagnosis frequency histogram, was used to construct age frequency

density plots by 1-year age increments [11,12]. Briefly, the vertical axis for each density plot represented the proportion of cases diagnosed with breast carcinoma at the corresponding age-at-diagnosis on the horizontal axis.

Survival and mortality

The Kaplan–Meier product-limit method was used to estimate breast cancer-specific survival, stratified by major ethnic groups [13]. Survival curves for each ethnic group were compared with the log rank test [14]. Percent annual breast cancer-specific hazard (or mortality) function was calculated with SEER Stat 6.1.4 using expected survival to compute the crude probability of death [15]. The hazard function is defined as a conditional rate of breast cancer-specific death at a given time interval following breast cancer diagnosis, provided that the subject was alive at the beginning of the time interval.

Results

Incidence rates and rate ratios (RR)

In the continental US, median ages at diagnosis ranged from a high of 62 years for White women to a low of 54 years for API women (Table 1). Median tumor size ranged from 2.0 cm for Black women to 1.5 cm for White women. In Hawaii, median ages were 59 and 61 years for Whites and APIs, respectively. Median tumor sizes were 1.5 cm for both groups.

Overall age-adjusted incidence rates ranged from a high of 178.1 per 100,000 woman-years for White women in Hawaii to a low of 104.1 per 100,000 woman-years for API women in the continental US. Incidence rates were higher in Hawaii than in the continental US among both White and API racial groups. For example, the incidence rate for Whites in Hawaii was 7.5% higher than for Whites in the continental US (178.1 compared to 165.6 per 100,000 woman-years). More strikingly, the incidence rate for APIs in Hawaii was 41% higher than for APIs in the continental US (147.2 compared to 104.1 per 100,000 woman-years).

Duct carcinoma not otherwise specified was the most common histopathologic subtype among all ethnic groups, ranging from a high of 76% for APIs in Hawaii to a low of 64% for Whites in the continental US. Compared to the White referent group in the continental US, duct NOS and tubular carcinomas were most common among Whites in Hawaii (RR=1.2 and 1.9), respectively. On the other hand, lobular and comedo-carcinomas were most common among Whites in the continental US. Blacks had the highest rates for medullary (RR=2.0) and inflammatory (RR=1.4) breast carcinomas. Papillary tumors were most common for Blacks and APIs in Hawaii, i.e., RR=1.5 for both groups. Notably, mucinous tumors were most common among APIs in Hawaii (RR=1.2) and least common among continental US APIs (RR=0.8).

Table 1. Female *in situ* and invasive breast cancer cases ($n = 323,607$) from SEER's 13 Registry Database, diagnosed during the years 1992–2002

Continental US (<i>n</i> = 313,036)													Hawaii (<i>n</i> = 10,571)															
Variable	White (referent)					Black					Asian/Pacific Islander					White					Asian/Pacific Islander							
	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI			
<i>Age at diagnosis</i>																												
< 50	59,160	22	53.1	1.0		8594	31	51.3	1.0		6070	37	45.1	0.9		766	26	58.5	1.1		1723	23	52.6	1.0		LB	UB	
50–59	58,453	22	361.6	1.0		6574	24	316.4	0.9		4382	27	254.4	0.7		758	25	405.6	1.1		1755	24	350.4	1.0		1.0	0.9	1.0
60–69	57,625	22	498.3	1.0		5587	20	409.3	0.8		3233	20	268.4	0.5		580	19	491.2	1.0		1867	25	452.1	0.9		0.9	0.9	1.0
70–79	57,654	22	579.7	1.0		4493	16	476.8	0.8		2114	13	272.5	0.5		626	21	646.3	1.1		1541	21	459.4	0.8		0.8	0.8	0.8
80+	32,841	12	495.1	1.0		2164	8	413.9	0.8		719	4	224.7	0.5		265	9	503.7	1.0		504	7	305.3	0.6		0.6	0.6	0.7
<i>Tumor size</i>																												
≤ 2.0 cm	146,095	55	91.2	1.0		11,627	42	61.6	0.7		8507	52	54.1	0.6		1747	58	104.0	1.1		4525	61	89.5	1.0		1.0	1.0	1.0
> 2.0 cm	71,552	27	44.4	1.0		9988	36	51.7	1.2		5377	33	33.4	0.8		774	26	45.8	1.0		1883	25	38.1	0.9		0.8	0.9	0.9
Other or unknown	48,086	18	–	–		5797	21	–	–		2634	16	–	–		474	16	–	–		982	13	–	–		–	–	–
<i>Lymph nodes</i>																												
LN negative	179,816	68	112.4	1.0		16,197	59	85.7	0.8		11,250	68	71.2	0.6		2,151	72	128.1	1.1		5544	75	109.9	1.0		1.0	1.0	1.0
LN positive	62,301	23	39.3	1.0		7954	29	40.6	1.0		4224	26	25.8	0.7		627	21	36.9	0.9		1481	20	30.0	0.8		0.7	0.8	0.8
Other or unknown	23,616	9	–	–		3261	12	–	–		1044	6	–	–		217	7	–	–		365	5	–	–		–	–	–
<i>AJCC staging</i>																												
Stage 0	43,216	16	27.5	1.0		4482	16	23.8	0.9		3165	19	19.8	0.7		485	16	28.7	1.0		1497	20	30.0	1.1		1.0	1.0	1.1
Stage I	99,516	37	62.0	1.0		7130	26	38.0	0.6		5517	33	35.2	0.6		1,221	41	72.8	1.2		3070	42	60.5	1.0		0.9	1.0	1.0
Stage II	74,353	28	46.6	1.0		8684	32	44.5	1.0		5243	32	32.4	0.7		829	28	49.0	1.1		1919	26	38.8	0.8		0.8	0.9	0.9
Stage III	13,397	5	8.3	1.0		2315	8	11.9	1.4		886	5	5.5	0.7		117	4	6.9	0.8		287	4	5.8	0.7		0.6	0.8	0.8
Stage IV	9024	3	5.6	1.0		1586	6	8.4	1.5		493	3	3.1	0.6		93	3	5.5	1.0		219	3	4.4	0.8		0.7	0.9	0.9
Other or unknown	26,227	10	–	–		3215	12	–	–		1214	7	–	–		250	8	–	–		398	5	–	–		–	–	–
<i>Tumor grade</i>																												
Low grade	125,350	47	77.9	1.0		9670	35	51.9	0.7		7566	46	48.4	0.6		1,512	50	90.2	1.2		3855	52	76.4	1.0		1.0	0.9	1.0
High grade	79,609	30	50.1	1.0		10,674	39	54.3	1.1		5840	35	35.8	0.7		797	27	47.0	0.9		2066	28	41.8	0.8		0.8	0.8	0.9
Other or unknown	60,774	23	–	–		7068	26	–	–		3112	19	–	–		686	23	–	–		1469	20	–	–		–	–	–

Table 1. Continued

Variable	Continental US (<i>n</i> = 313,036)										Hawaii (<i>n</i> = 10,571)																		
	White (referent)					Black					Asian/Pacific Islander					White					Asian/Pacific Islander								
	<i>n</i>	%	Rate	RR	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI	<i>n</i>	%	Rate	RR	95% CI
<i>Age at diagnosis</i>																													
<i>Histologic type</i>																													
Duct NOS	171,189	64	106.9	1.0	18,022	66	94.0	0.9	0.9	11,188	68	70.0	0.7	0.6	0.7	2,163	72	128.4	1.2	1.2	1.3	5616	76	112.1	1.0	1.0	1.1		
Tubular	3958	1	2.5	1.0	179	1	1.0	0.4	0.3	93	1	0.6	0.2	0.2	0.3	78	3	4.6	1.9	1.5	2.3	80	1	1.6	0.6	0.5	0.8		
Lobular	24,862	9	15.5	1.0	1657	6	8.9	0.6	0.5	752	5	4.8	0.3	0.3	0.3	208	7	12.4	0.8	0.7	0.9	273	4	5.5	0.4	0.3	0.4		
Medullary	1881	1	1.2	1.0	498	2	2.4	2.0	1.8	153	1	0.9	0.8	0.7	0.9	18	1	1.0	0.9	0.5	1.4	42	1	0.9	0.7	0.5	1.0		
Inflammatory	2388	1	1.5	1.0	414	2	2.1	1.4	1.3	129	1	0.8	0.5	0.4	0.6	15	1	0.9	0.6	0.3	1.0	32	0	0.6	0.4	0.3	0.6		
Papillary	3548	1	2.2	1.0	572	2	3.2	1.5	1.3	346	2	2.3	1.0	0.9	1.2	28	1	1.7	0.8	0.5	1.1	168	2	3.3	1.5	1.3	1.8		
Mucinous	5634	2	3.4	1.0	519	2	3.0	0.9	0.8	429	3	2.8	0.8	0.7	0.9	58	2	3.5	1.0	0.8	1.4	206	3	4.0	1.2	1.0	1.4		
Paget's disease	1829	1	1.1	1.0	195	1	1.0	0.9	0.8	120	1	0.8	0.7	0.6	0.9	24	1	1.4	1.3	0.8	1.9	36	0	0.7	0.6	0.4	0.9		
Comedo	11,167	4%	7.1	1.0	1191	4	6.1	0.9	0.8	927	6	5.7	0.8	0.7	0.9	90	3	5.3	0.7	0.6	0.9	224	3	4.5	0.6	0.6	0.7		
Other or unknown	39,277	15	—	—	4165	15	—	—	—	2381	14	—	—	—	—	313	10	—	—	—	—	713	10	—	—	—	—	—	
<i>ER</i>																													
ER positive	143,512	54	89.1	1.0	10,728	39	57.2	0.6	0.6	8078	49	51.1	0.6	0.6	0.6	1,759	59	105.0	1.2	1.2	1.2	4415	60	87.5	1.0	1.0	1.0		
ER negative	39,117	15	24.8	1.0	6533	24	32.9	1.3	1.3	2727	17	16.7	0.7	0.6	0.7	534	18	31.4	1.3	1.2	1.4	1168	16	23.6	1.0	0.9	1.0		
Other or unknown	83,104	31	—	—	10,151	37	—	—	—	5713	35	—	—	—	—	702	23	—	—	—	—	1807	24	—	—	—	—	—	
<i>PR</i>																													
PR positive	119,803	45	74.6	1.0	8785	32	46.4	0.6	0.6	6850	41	43.0	0.6	0.6	0.6	1,550	52	92.3	1.2	1.2	1.3	3998	54	79.8	1.1	1.0	1.1		
PR negative	56,773	21	35.5	1.0	7816	29	40.1	1.1	1.0	3532	21	22.1	0.6	0.6	0.6	725	24	43.0	1.2	1.1	1.3	1527	21	30.3	0.9	0.8	0.9		
Other or unknown	89,157	34	—	—	10,811	39	—	—	—	6136	37	—	—	—	—	720	24	—	—	—	—	1865	25	—	—	—	—	—	

Key: yrs, years; cm, centimeters; Rate, incidence rate per 100,000 woman-years, age-adjusted (2000 US standard); RR, rate ratio where a given characteristic is compared to the referent value in the Continental US White race with an assigned RR of 1.0; 95% CI, 95% confidence intervals with lower bounds (LB) and upper bounds (UB), Duct NOS, ductal carcinoma not otherwise specified; ER, estrogen receptor; PR, progesterone receptor.

Table 2. Racial groupings for *in situ* and invasive female breast cancer cases ($n=323,607$) from SEER's 13 Registry Database in Hawaii and in the Continental United States

Race	Continental US ($n=313,036$)		Hawaii Registry ($n=10,571$)	
	<i>n</i>	% Total	<i>n</i>	% Total
<i>Major racial groups</i>				
White	265,733	84.89	2995	28.33
White NOS	243,839		2977	
White, Hispanic Origin	21,894		18	
Black	27,412	8.76	56	0.53
American Indian/Alaska Native	1434	0.46	28	0.26
Asian or Pacific Islander (API)	16,518	5.28	7390	69.91
Other or unknown	1939	0.62	102	0.96
Total	313,036	100.00	10,571	100.00
<i>Asian and Pacific Islander (API) subgroups</i>				
Asian subgroups	<i>n</i>	% API	<i>n</i>	% API
Chinese	4420	26.76	698	9.45
Japanese	2592	15.69	3614	48.90
Filipino	4458	26.99	1059	14.33
Korean (1988+)	948	5.74	198	2.68
Asian Indian, Pakistani (1988+)	864	5.23	11	0.15
Vietnamese (1988+)	879	5.32	27	0.37
Laotian (1988+)	41	0.25	3	0.04
Hmong (1988+)	2	0.01	0	0.00
Kampuchean (1988+)	73	0.44	0	0.00
Thai (1994+)	169	1.02	5	0.07
Chamorroan (1991+)	2	0.01	1	0.01
Other Asian (1991+)	1414	8.56	39	0.53
Asian subtotal	15,862	96.03	5655	76.52
Pacific Islander subgroups	<i>n</i>	% API	<i>n</i>	% API
Hawaiian	140	0.85	1620	21.92
Micronesian, NOS (1991+)	7	0.04	17	0.23
Guamanian, NOS (1991+)	19	0.12	4	0.05
Polynesian, NOS (1991+)	12	0.07	1	0.01
Tahitian (1991+)	3	0.02	0	0.00
Samoan (1991+)	93	0.56	68	0.92
Tongan (1991+)	43	0.26	18	0.24
Melanesian, NOS (1991+)	7	0.04	0	0.00
Fiji Islander (1991+)	33	0.20	0	0.00
New Guinean (1991+)	1	0.01	0	0.00
Pacific Islander, NOS (1991+)	38	0.23	6	0.08
Pacific Islander subtotal	396	2.40	1734	23.46
Other Asian and Pacific Islander	260	1.57	1	0.01
Asian and Pacific Islander (total)	16,518	100.00	7390	100.00

Key: NOS, not otherwise specified.

In the continental US, incidence rates were most favorable for APIs and least favorable for Blacks. For example, RRs for large (> 2.0 cm) tumor diameter were 0.8 and 1.2 for APIs and Blacks, respectively. We observed similar patterns for AJCC stage, grade, ER and PR expression. Lower incidence rates for APIs compared to Whites in the continental US were especially pronounced at ages greater than 60 years ($RR=0.5$) as well as for certain histologic types such as tubular ($RR=0.2$), lobular ($RR=0.3$), and inflammatory ($RR=0.5$) breast

carcinomas. Differences between APIs and Whites were less obvious at ages < 50 years ($RR=0.9$). Higher incidence rates for Blacks compared to Whites in the continental US were most evident for AJCC stage III ($RR=1.4$) and stage IV ($RR=1.5$) as well as for medullary ($RR=2.0$), inflammatory ($RR=1.4$), and papillary ($RR=1.5$) carcinomas.

In Hawaii as in the continental US, incidence rates were more favorable for APIs than for Whites. For example, RR for large (> 2.0 cm) tumor diameter was

0.9 and 1.0 for APIs and Whites in Hawaii, respectively. We observed similar trends for nodal status, AJCC stage, grade, ER and PR expression. In Hawaii, as in the continental US, lower incidence rates for APIs were more pronounced at older ages (RR=0.6 for ages 80+ years) as well as for certain histologies such as lobular (RR=0.4) and inflammatory (RR=0.4) carcinomas. Higher incidence rates for Whites in Hawaii were most evident for tubular carcinomas (RR=1.9), ER negative (RR=1.3) and PR negative (RR=1.2) expression.

Age distribution at diagnosis

Figure 1 illustrates the age distributions at diagnosis for breast carcinoma cases by ethnic group, with age-specific incidence rates shown in Figure 1 (a) and age frequency distributions in Figure 1(b–d). Under age 40 years, age-specific rates for Black women in the continental US were higher than other groups. After age 50 years, incidence rates increased at a slower rate for all racial groups. Between ages 50–70 years, rates for APIs in Hawaii were higher than those for Blacks. At older ages, incidence rates could be ranked in decreasing order from Whites to Blacks to APIs.

Age frequency distributions for Whites in the continental US and Whites in Hawaii differed (Figure 1b). The age distribution for Whites in the continental US was bimodal with early- and late-onset peak frequencies at approximately ages 50 and 70 years, respectively. On the other hand, Whites in Hawaii had a predominant early-onset peak frequency at age 50 years. The Blacks in the continental US also had an early-onset mode at age 50 years (Figure 1c). Finally, age frequency distributions for APIs in Hawaii and in the continental US were dissimilar. That is, the age distribution for APIs in Hawaii was bimodal (Figure 1d), whereas the age distribution for continental US APIs had a single early-onset mode at approximately age 50 years.

Survival and mortality

Breast cancer-specific survival curves are depicted in Figure 2(a). Cancer-specific survival for White women in Hawaii was slightly better than for White women in the continental US (log rank test, $p=0.036$). Similarly, cancer-specific survival for API women in Hawaii was better than for API women in the continental US ($p=0.003$). Cancer-specific survival for Blacks in the

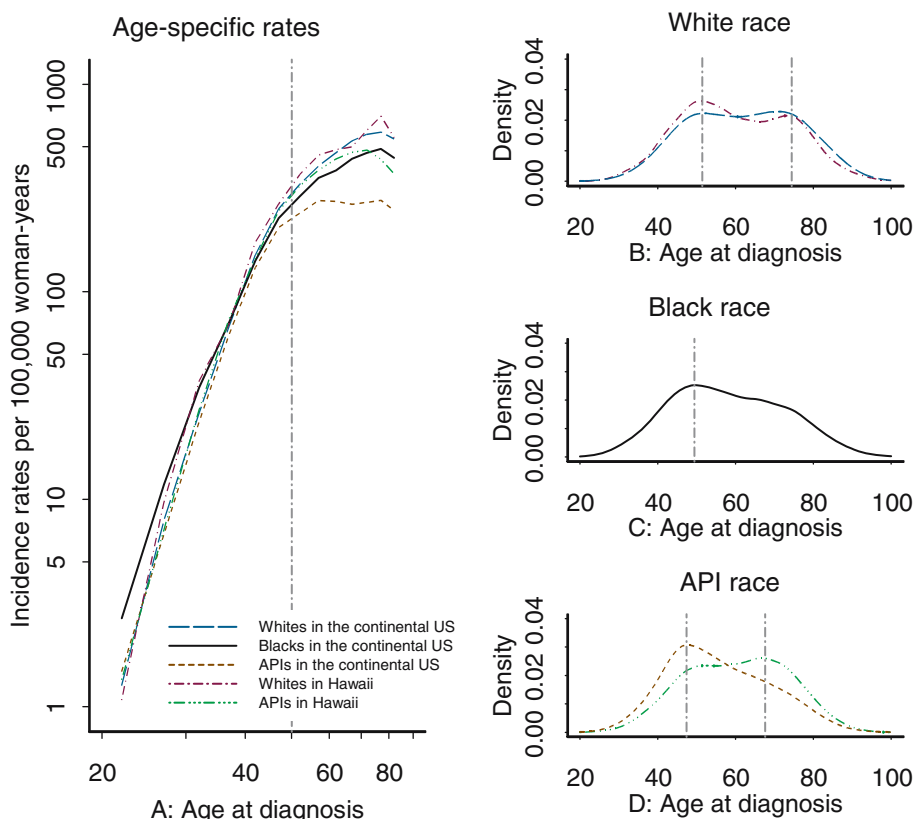


Figure 1. Age distribution (rates and frequency) for female breast cancer cases among ethnic groups in the continental United States (US) and in the state of Hawaii. Breast cancer cases were obtained from the Surveillance, Epidemiology, and End Results (SEER) 13 Registry, which were diagnosed during the years 1992–2002. (a) Includes five age-specific incidence rate curves for: (i) White race in the continental US (blue line); (ii) Black race in the continental US (black line); (iii) API race in the continental US (orange line); (iv) White race in Hawaii (red line); (v) Asian or Pacific Islander (API) race in Hawaii (green line). (b) Includes two age density plots for: (i) White race in the continental US (blue line); (ii) White race in Hawaii (red line). (c) Includes one age density plot for Black race in the continental US (black line). (d) Includes two age density plots for: (i) API race in the continental US (orange line); (ii) API race in Hawaii (green line).

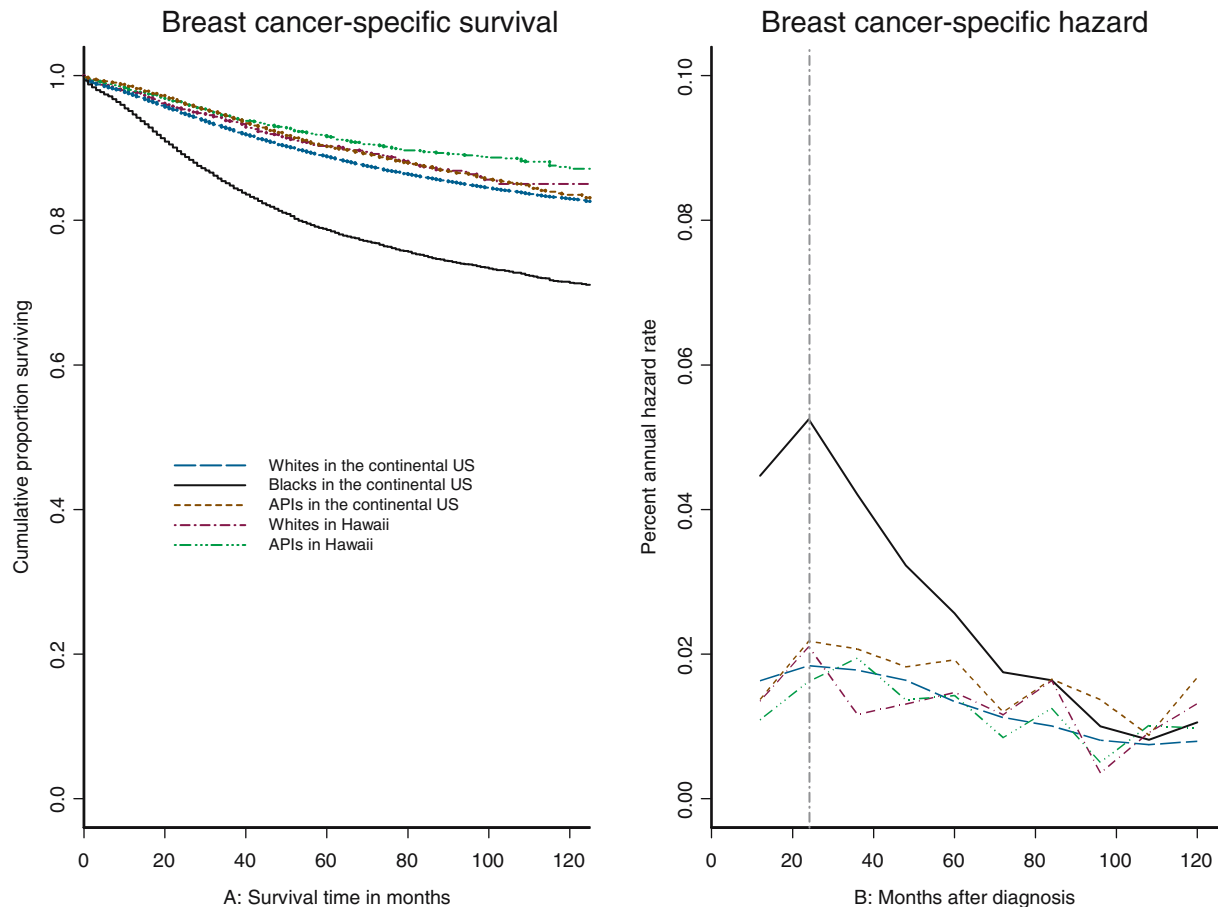


Figure 2. Breast cancer-specific survival (a) and hazard function (b) for female breast cancer cases among ethnic groups in the continental United States (US) and in the state of Hawaii. Breast cancer cases were obtained from the Surveillance, Epidemiology, and End Results (SEER) 13 Registry, which were diagnosed during the years 1992–2002. (i) White race in the continental US (blue line); (ii) Black race in the continental US (black line); (iii) API race in the continental US (orange line); (iv) White race in Hawaii (red line); (v) Asian or Pacific Islander (API) race in Hawaii (green line).

continental US was markedly worse than for all other racial groups (log rank test, $p < 0.0001$).

Figure 2b demonstrates that the greatest risk of breast cancer-specific death occurred approximately 2–3 years for Blacks. Indeed, annual breast cancer-specific mortality peaked at 5.25% per year among Black women, approximately 24 months after breast cancer diagnosis. The excess annual hazard rate among Black women fell for the next 80 months, after which, the annual risk for breast cancer death was similar for all racial groups.

Discussion

This analysis of SEER data for $n = 323,607$ *in situ* and invasive breast carcinoma cases identified *inter-* and *intra-*ethnic disparities (i.e., between and within racial groups) in the continental US and in the state of Hawaii, including different age distributions at diagnosis (rates and frequency), tumor profiles, histopathologic subtypes, cancer-specific survival and mortality hazard function. Incidence tumor characteristics and prognostic patterns were most favorable for APIs in the continental

US and worse for Blacks in the continental US. In general, Whites in the continental US had intermediate risk and prognostic profiles.

In the continental US, age-specific incidence rates for White and Black women followed a pattern that has been well-established [12,16]. Rates for Whites increased rapidly until age 50 years then continued to rise at a slower rate. Rates were higher for Blacks than for Whites until ages 40–44 years, after which there was ethnic crossover with rates higher for Whites than for Blacks [17]. Rates for APIs in the continental US increased rapidly until age 50 years then flattened, similar to patterns in low risk countries [18].

Divergent age-specific incidence rate patterns for different racial groups in the continental US reflected different breast cancer populations, as demonstrated with density plots. For example, Whites in the continental US had bimodal breast cancer populations with a predominant late onset mode; whereas Blacks and APIs in the continental US had predominantly early-onset modes of approximately age 50 years. In Hawaii, density plots for Whites demonstrated a predominant early-onset component, i.e., similar to Blacks in the continental

US; whereas late-onset tumors dominated for APIs in Hawaii, i.e., similar to Whites in the continental US.

The *inter*-ethnic differences between racial groups in this study were expected and may be the product of non-biologic as well as biologic factors. For example, *inter*-racial disparity for tumor size, nodal status and AJCC stage at diagnosis are often attributed to non-biologic factors such as access to and/or utilization of medical care [19,20]. However, if denied or delayed access to health care were the sole determinant for worse or later stage at diagnosis for Blacks compared to all other racial groups, Black women should develop breast tumors later (not earlier) in life. In other words, delayed diagnosis should be associated with older (not earlier) ages at diagnosis [21,22]. So, biologic determinants may also play a role for *inter*-racial disparity [3,23–26].

In contrast to the expected *inter*-ethnic differences between racial groups, *intra*-ethnic differences within major racial groupings were somewhat unexpected, but probably reflected fundamental differences within SEER's major racial groupings. For example, White women in the continental US included 8.2% Caucasians of Hispanic origin ($n=21,894$ of 265,733) compared to less than 1% Hispanic origin ($n=18$ of 2995) in Hawaii (Table 2). Given that Hispanics have lower breast cancer risk than non-Hispanic Whites [27], Hispanics in the continental US may have reduced overall rates among Whites in the continental US compared to Whites in Hawaii. API women with breast carcinoma in the continental US included 96.03% Asians (27% Chinese, 16% Japanese, and 27% Filipino) and 2.4% Pacific Islanders. In contrast, API women with breast carcinoma in Hawaii included 76.5% Asians (9% Chinese, 49% Japanese, and 14% Filipino) and 23.46% Pacific Islanders. In accordance with their low-risk incidence rate pattern [28], API women in the continental US also appeared to be largely first-generation migrants with less than 8% (1,313 of 16,518) recorded as born in the continental US. On the other hand and consistent with their high-risk 'westernized' incidence rate pattern, API women in Hawaii were mostly subsequent-generation immigrants with 66% (4867 of 7390) listed as born in the state of Hawaii. Other potentially important variables such as socioeconomic status or educational level were not recorded in the SEER database.

In sum, we observed *inter*- and *intra*-ethnic differences for female breast carcinoma for White and API races in the continental US and the state of Hawaii. While the *inter*-racial variations were consistent with previous studies, the *intra*-racial disparities may reflect geographic variations for heterogeneous non-biologic and biologic factors among racial subgroups in SEER, which were combined into larger groups and then analyzed together. Future studies should attempt to disaggregate data to separately characterize incidence and prognostic patterns for racial groupings. Accomplishing this goal will require the NCI's continued commitment for large-scale population-based data sets with ethnic diversity. Over

sampling for some racial subgroups may be needed to ensure adequate representation within SEER. This effort is warranted given the implications for prevention, early detection, and treatment of breast carcinoma for the special populations in the United States.

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